

E13 Multiple reactions in a non-isothermal batch reactor

Sunday, January 14, 2024

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The following reactions are taking place in the liquid phase in a 2000-L batch reactor:

$A + 2B \rightarrow C$	$\Delta H_{rxn} = -5 \text{ kcal/mol B}$	$-r_{1A} = k_{1A} \cdot C_A \cdot C_B^2$
$3C + 2A \rightarrow D$	$\Delta H_{rxn} = +10 \text{ kcal/mol C}$	$-r_{2A} = k_{2A} \cdot C_A \cdot C_C$
$B + 3C \rightarrow E$	$\Delta H_{rxn} = -50 \text{ kcal/mol B}$	$-r_{3C} = k_{3C} \cdot C_B \cdot C_C$

$$\frac{r_{1A}}{-1} = \frac{r_{1B}}{-2} \quad , \quad r_{1B} = 2 r_{1A}$$

The initial temperature is 450 K and the initial concentrations of A, B, and C are 1.0, 0.5, and 0.2 M, respectively. The coolant temperature is constant at 400 K, and $UA = 0.1 \text{ kcal/s} \cdot \text{K}$. Given the following information, plot C_i and T as a function of time for up to 1 hr.

Data:

$$k_{1A} = 1e-3 \text{ M}^{-2} \cdot \text{s}^{-1}$$

$$k_{2A} = (1/3)e-3 \text{ M}^{-1} \cdot \text{s}^{-1}$$

$$k_{3C} = 0.6e-3 \text{ M}^{-1} \cdot \text{s}^{-1}$$

All rate constant values given at 450 K

$$CP_A = 0.01 \text{ kcal/mol/K}$$

$$CP_B = 0.01 \text{ kcal/mol/K}$$

$$CP_C = 0.05 \text{ kcal/mol/K}$$

$$CP_D = 0.08 \text{ kcal/mol/K}$$

$$CP_E = 0.05 \text{ kcal/mol/K}$$

rate balances:

$$\frac{dC_A}{dt} = r_A \quad \frac{dC_B}{dt} = r_B \quad \frac{dC_C}{dt} = r_C$$

$$\frac{dC_D}{dt} = r_D \quad \frac{dC_E}{dt} = r_E$$

net rates of generation

$$r_A = r_{1A} + r_{2A}$$

$$r_B = r_{1B} + r_{3E}$$

$$r_C = r_{1C} + r_{2C} + r_{3C}$$

$$r_D = r_{2D}$$

$$r_E = r_{3E}$$

relative rates

$$r_{1B} = 2r_{1A}$$

$$r_{1C} = -r_{1A}$$

$$r_{2C} = \frac{3}{2} r_{2A}$$

$$r_{2D} = -2 r_{2A}$$

$$r_{3B} = \frac{1}{3} r_{3C}$$

$$r_{3E} = -r_{3C}$$

energy balance

$$\frac{dT}{dt} = \frac{V \left[(\Delta H_{rxn1B})(r_{1B}) + (\Delta H_{rxn2C})(r_{2C}) + (\Delta H_{rxn3E})(r_{3E}) \right] - UA(T - T_a)}{N_A C_{pA} + N_B C_{pB} + N_C C_{pC} + N_D C_{pD} + N_E C_{pE}}$$

equations for N_i

$$N_A = C_A V$$

$$N_B = C_B V$$

$$N_C = C_C V$$

$$N_D = C_D V$$

$$N_E = C_E V$$

Arrhenius equations

$$k_{1A} = k_{1A0} \exp \left(\frac{E_{a1}}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right)$$

$$k_{2A} = k_{2A0} \exp \left(\frac{E_{a2}}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right)$$

$$k_{3c} = k_{3c0} \exp \left(\frac{E_{a3}}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right)$$